

Solve each problem, then find your answer in the DECODER and write its letter above that problem's number at the bottom of page 2.

Self-check: if your answer is not in the decoder, you have a mistake — go back and find it. When you finish, the letters spell the answer to this:

What do you call an angle after a car crash?

DECODER

$(-2, 7) = W$	$(4, 6) = A$	$(6, 4) = B$	$(12, -6) = R$	$5\sqrt{2} = K$	NONAGON = L
$6 = E$	$8 = F$	$10 = C$	$13 = H$	$24 = P$	$28 = D$
$30 = N$	$34 = E$	$36 = G$	$48 = A$	$56 = E$	$60 = Y$

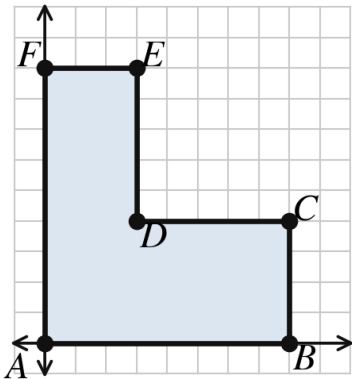
Part A — 1.3 Midpoint and Distance

- Find the midpoint of $\overline{AB}$ with endpoints $A(-4, 3)$ and $B(12, 9)$.
- Find the midpoint of $\overline{CD}$ with endpoints $C(-7, 2)$ and $D(3, 12)$.
- $M(5, -1)$ is the midpoint of $\overline{RS}$. R has coordinates $(-2, 4)$. Find the coordinates of S .
- C is the midpoint of $\overline{DE}$. $DC = 3x - 4$ and $CE = x + 8$. Find x .
- Use the Distance Formula to find PQ . $P(2, -3)$ and $Q(10, 3)$
- Use the Distance Formula to find GH . Simplify. $G(-5, 4)$ and $H(0, -1)$
- Line m bisects $\overline{JK}$ at P . $JP = 5x + 3$ and $PK = 8x - 12$. Find JK .

Part B — 1.4 Polygons, Perimeter and Area

8. The vertices of rectangle JKLM are J(−3, 4), K(5, 4), L(5, −2), M(−3, −2). Find the perimeter of JKLM.
9. Find the area of rectangle JKLM from problem 8.
10. Find the perimeter of $\triangle FGH$ with F(−2, −1), G(10, −1), and H(10, 4).

11. Find the area of $\triangle PQR$ with P(1, 2), Q(9, 2), and R(9, 11).
12. What is the name of a polygon with 9 sides?
13. Find the perimeter of the figure at right.



D E C O D E R					
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6 = E	8 = F	10 = C	13 = H	24 = P	28 = D
30 = N	34 = E	36 = G	48 = A	56 = E	60 = Y

What do you call an angle after a car crash?

1 2 3 4 5 6 7 8 9 10 11 12 13

Doesn't spell anything? One of your answers is wrong — hunt it down.